Worksheet 1.2 — Characteristics of Polynomial Functions

Grade 12 Advanced Functions (MHF4U) · Unit 1: Polynomial Functions

The Big Ideas

Degree n limits the graph: at most n zeros and $n-1$ turning points; the n-th finite differences are constant.

- At most n x-intercepts and $n-1$ turning points.
- The n-th differences of a degree-n polynomial are constant.
- Even: $f(-x) = f(x)$ (y-axis). Odd: $f(-x) = -f(x)$ (origin).

Core Concepts

Built from power functions

A **polynomial** is a sum of power functions; its degree still governs end behaviour and caps the number of intercepts and turning points.

Key features

Read a polynomial by its degree, leading coefficient, zeros (with multiplicity), and turning points at most $n-1$ for degree n.

Finite differences

The degree shows up in a table too: the nth finite differences of a degree-n polynomial are constant.

Worked Examples

Example 1 — Turning points

Max turning points of a degree-5 polynomial?

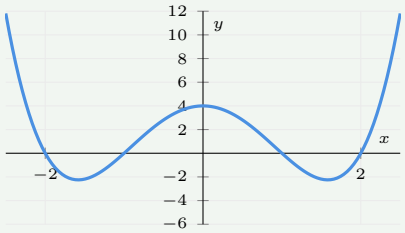
Solution. $n-1 = 5-1 = 4$. E.g. $y = x^5 - 5x^3 + 4x$ actually reaches that maximum:



Example 2 — x-intercepts

Max x-intercepts of a degree-4 polynomial?

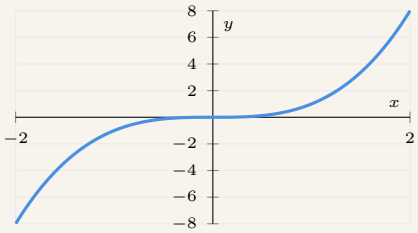
Solution. At most $n = 4$. E.g. $y = x^4 - 5x^2 + 4 = (x-1)(x+1)(x-2)(x+2)$ hits all 4:



Example 3 — Degree from differences

The 3rd finite differences are constant. Degree?

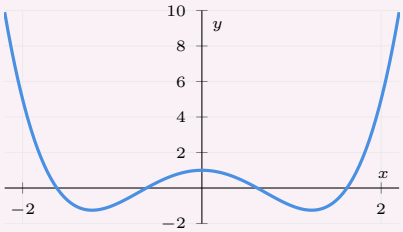
Solution. n-th differences constant ⇒ degree n, so 3 (a cubic shape):



Example 4 — Even function

Classify $f(x) = x^4 - 3x^2 + 1$.

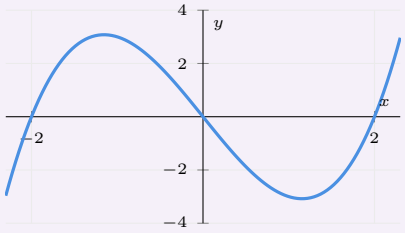
Solution. Only even powers ⇒ $f(-x) = f(x)$: even (y-axis symmetry):



Example 5 — Odd function

Classify $f(x) = x^3 - 4x$.

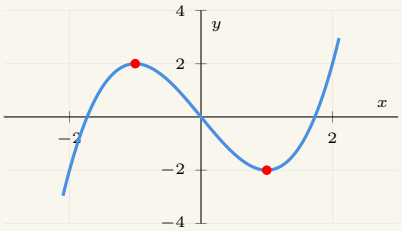
Solution. Only odd powers ⇒ $f(-x) = -f(x)$: odd (origin symmetry):



Example 6 — A cubic's graph

A cubic $y = x^3 - 3x$ has how many turning points and zeros?

Solution. 2 turning points (at $x = \pm 1$) and 3 zeros the maxima for degree 3:



Example 7 — Minimum degree

A polynomial has 4 turning points. Minimum degree?

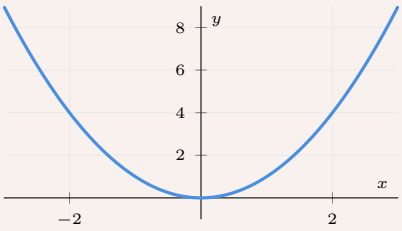
Solution. Turning points $\leq n-1$, so $n \geq 5$. E.g. $y = x^5 - 5x^3 + 4x$ (degree 5) reaches 4 turning points:



Example 8 — Degree from differences

The 2nd differences are constant. Degree?

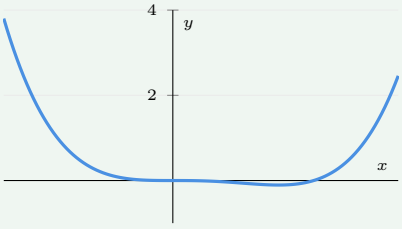
Solution. 2 (a parabola shape):



Example 9 — Neither

Classify $f(x) = x^4 - x^3$.

Solution. Mixed even and odd powers neither:



Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Max turning points of a degree-4 polynomial?

Question 2

Max x-intercepts of a degree-6 polynomial?

Question 3

The 4th differences are constant. Degree?

Question 4

Is $f(x) = x^6 - x^2$ even, odd, or neither?

Question 5

Is $f(x) = x^5 - x^3$ even, odd, or neither?

Question 6

Max turning points of a degree-7 polynomial?

Question 7

A polynomial has 5 turning points. Minimum degree?

Question 8

The 1st differences are constant. Degree?

Question 9

Is $f(x) = x^4 + x$ even, odd, or neither?

Question 10

Max x-intercepts of a degree-3 polynomial?

Question 11

Symmetry of an even function?

Question 12

Symmetry of an odd function?

Question 13 — Challenge

A degree-4 polynomial has exactly 1 turning point is that possible? Explain.

Question Recap & Answer Key

Question Recap

1. Max turning points of a degree-4 polynomial?
2. Max x-intercepts of a degree-6 polynomial?
3. The 4th differences are constant. Degree?
4. Is $f(x) = x^6 - x^2$ even, odd, or neither?
5. Is $f(x) = x^5 - x^3$ even, odd, or neither?
6. Max turning points of a degree-7 polynomial?
7. A polynomial has 5 turning points. Minimum degree?
8. The 1st differences are constant. Degree?
9. Is $f(x) = x^4 + x$ even, odd, or neither?
10. Max x-intercepts of a degree-3 polynomial?
11. Symmetry of an even function?
12. Symmetry of an odd function?
13. A degree-4 polynomial has exactly 1 turning point is that possible? Explain.

✓ Answer Key

1. 3
2. 6
3. 4
4. even
5. odd
6. 6
7. 6
8. 1
9. neither
10. 3
11. y-axis
12. origin
13. yes (it need not reach the maximum of 3)